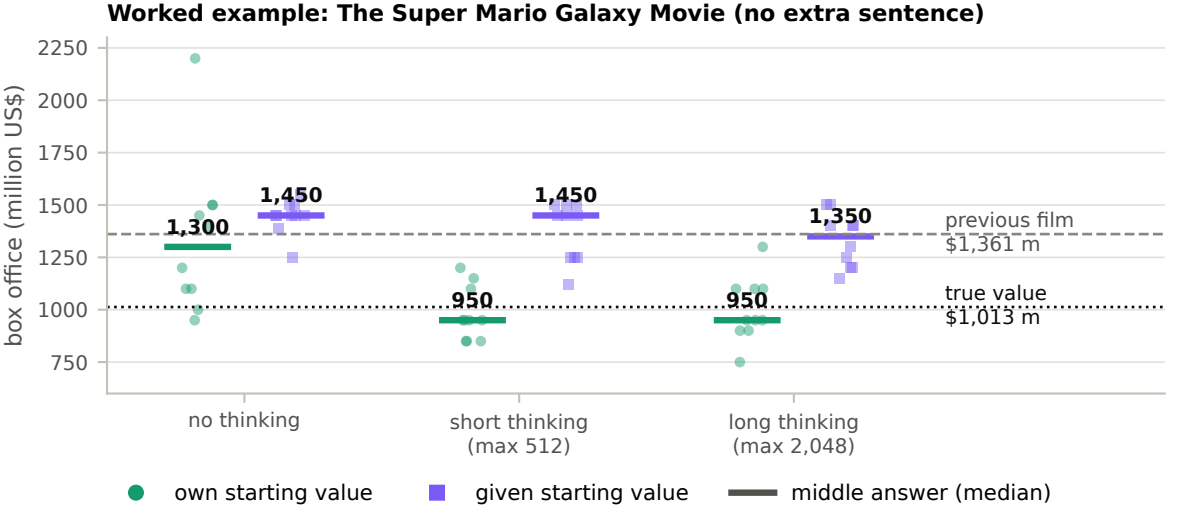
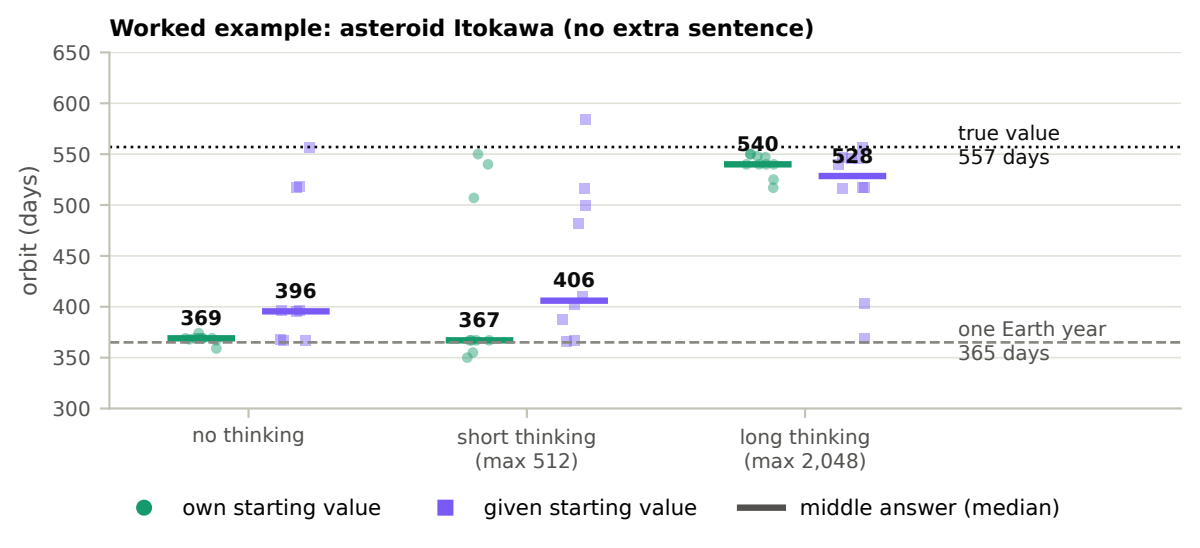
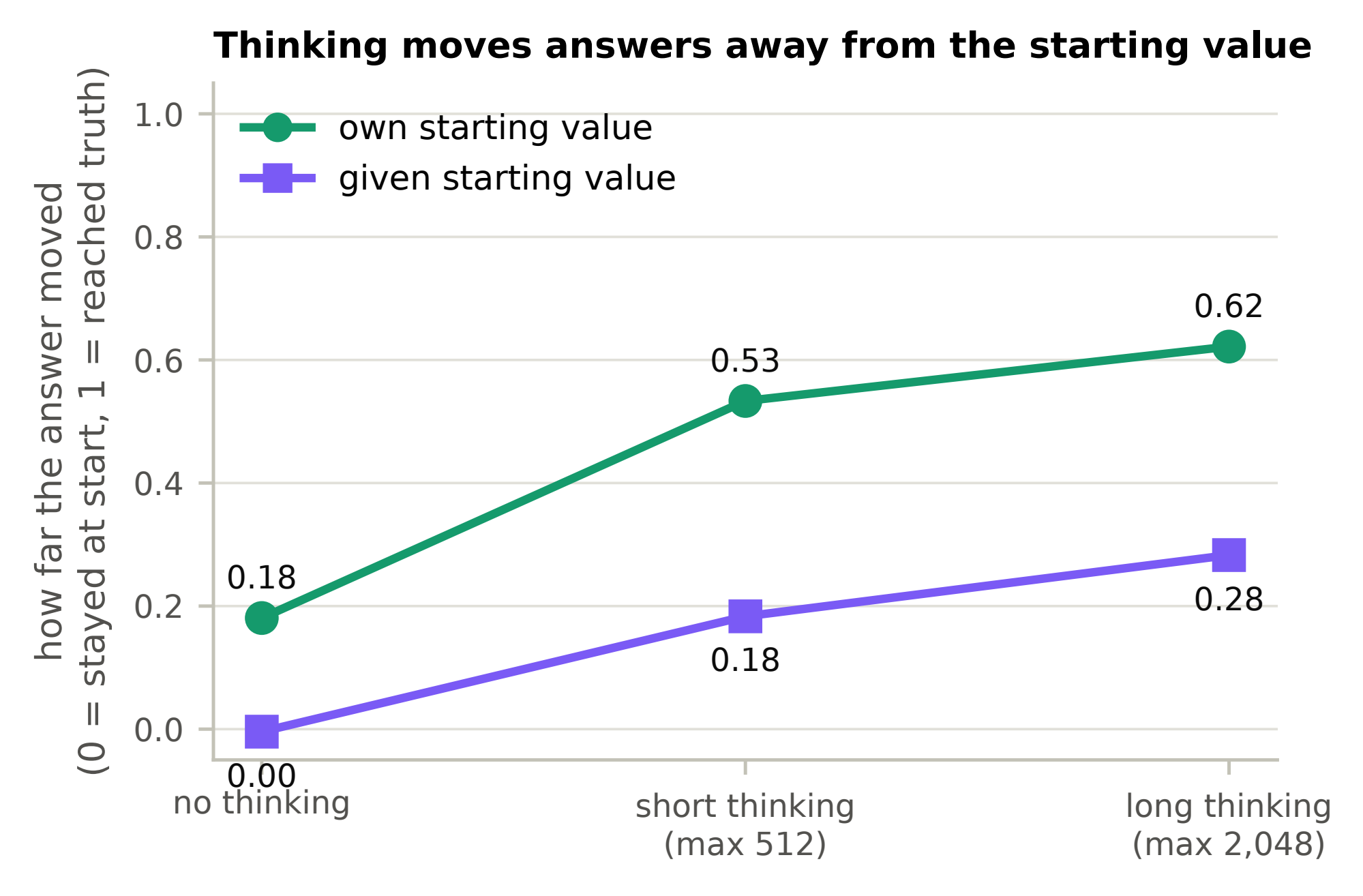
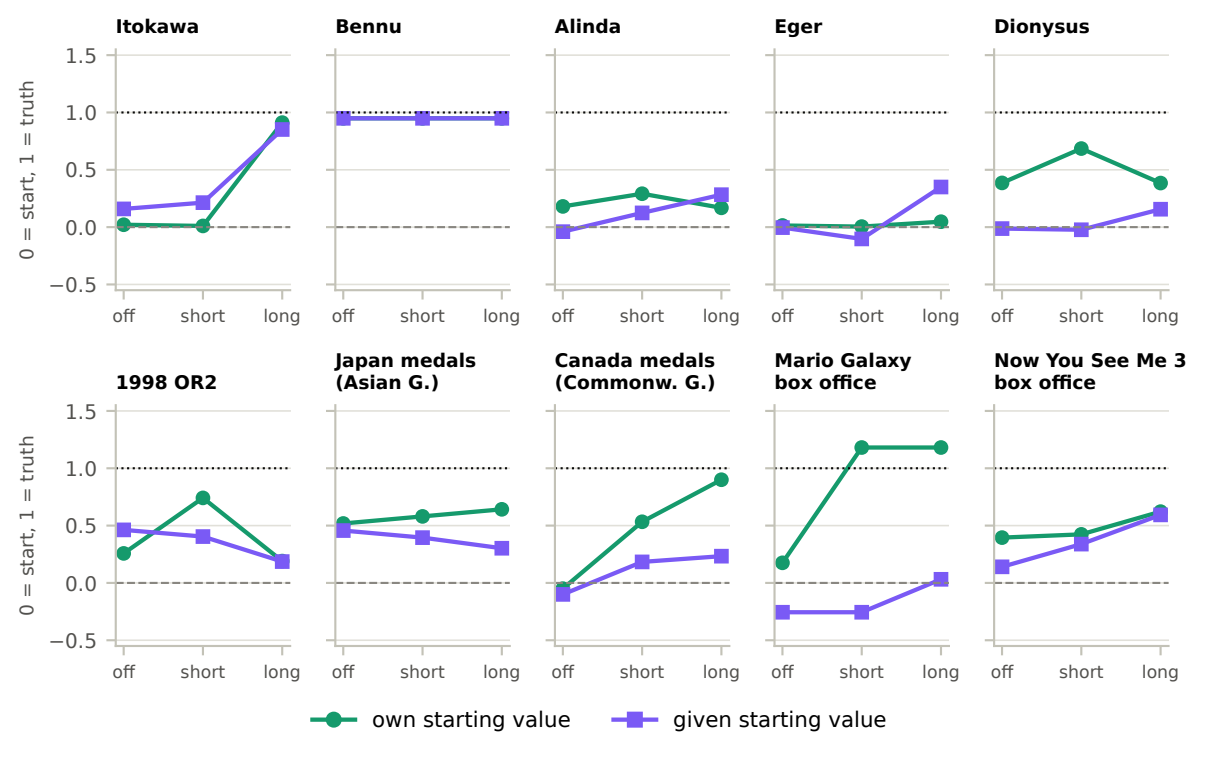
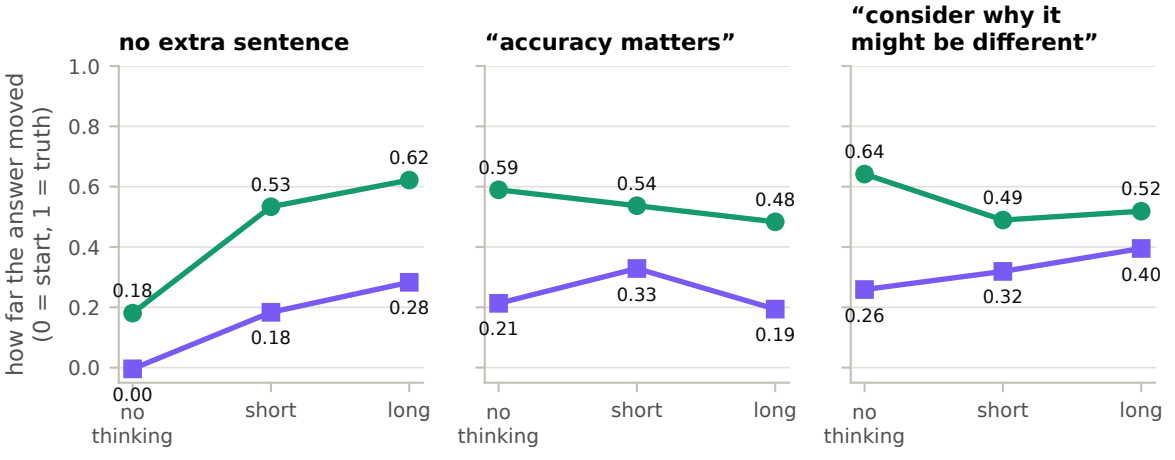
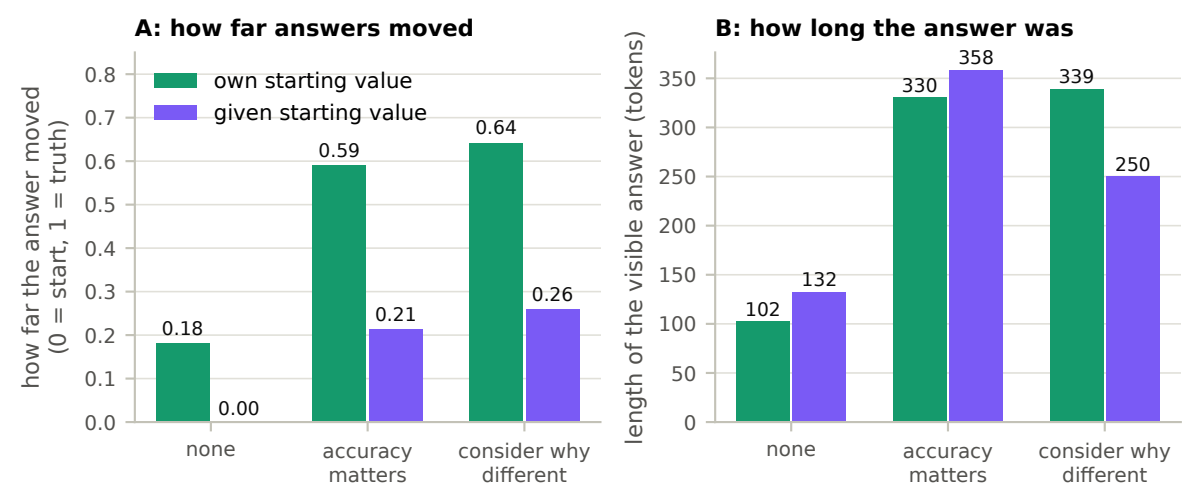

Worked example: The Super Mario Galaxy Movie (no extra sentence)
box office (million US$)
2250
2000
1750
1500
1250
1000
750
1,300
1,450
950
1,450
950
1,350
previous film
$1,361 m
true value
$1,013 m
no thinking
short thinking
(max 512)
long thinking
(max 2,048)
own starting value
given starting value
middle answer (median)

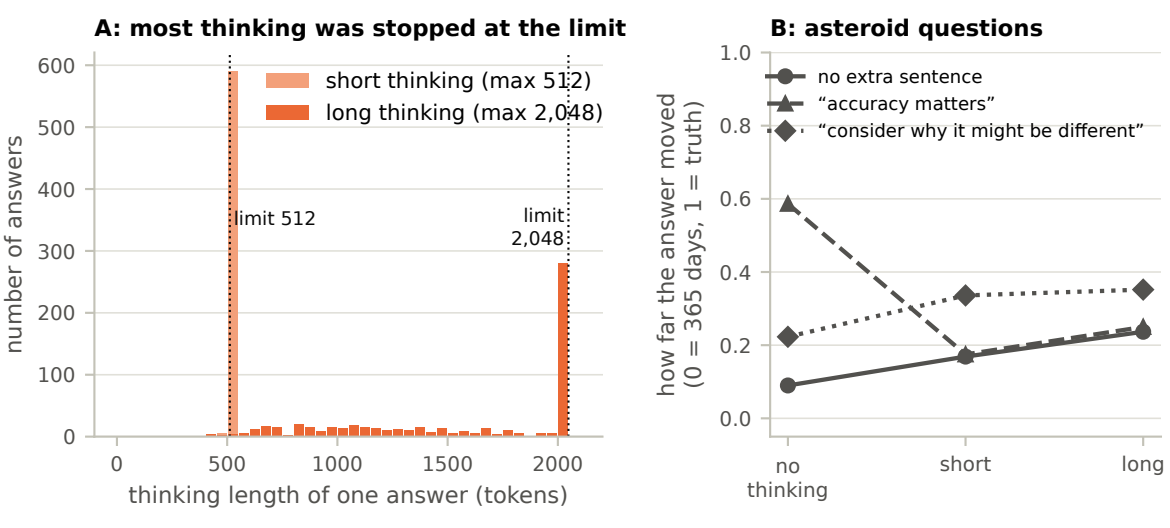

Worked example: asteroid Itokawa (no extra sentence)
650
600
550
500
450
400
350
300
orbit (days)
true value
557 days
one Earth year
365 days
369
396
367
406
540
528
no thinking
short thinking
(max 512)
long thinking
(max 2,048)
own starting value
given starting value
middle answer (median)


Thinking moves answers away from the starting value
how far the answer moved
(0 = stayed at start, 1 = reached truth)
1.0
0.8
0.6
0.4
0.2
0.0
own starting value
given starting value
0.18
0.53
0.62
0.00
0.18
0.28
no thinking
short thinking
(max 512)
long thinking
(max 2,048)


Itokawa
Bennu
Alinda
Eger
Dionysus
1998 OR2
Japan medals (Asian G.)
Canada medals (Commonw. G.)
Mario Galaxy box office
Now You See Me 3 box office
0 = start, 1 = truth
1.5
1.0
0.5
0.0
−0.5
off
short
long
own starting value
given starting value


no extra sentence
“accuracy matters”
“consider why it might be different”
how far the answer moved (0 = start, 1 = truth)
1.0
0.8
0.6
0.4
0.2
0.0
0.18
0.53
0.62
0.00
0.18
0.28
0.59
0.54
0.48
0.21
0.33
0.19
0.64
0.49
0.52
0.26
0.32
0.40
no thinking
short
long
own starting value
given starting value


A: how far answers moved
how far the answer moved
(0 = start, 1 = truth)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
own starting value
given starting value
0.18
0.00
0.59
0.21
0.64
0.26
none
accuracy matters
consider why different
B: how long the answer was
length of the visible answer (tokens)
0
50
100
150
200
250
300
350
102
132
330
358
339
250
none
accuracy matters
consider why different


A: most thinking was stopped at the limit
short thinking (max 512)
long thinking (max 2,048)
limit 512
limit 2,048
number of answers
thinking length of one answer (tokens)
0
100
200
300
400
500
600
0
500
1000
1500
2000
B: asteroid questions
no extra sentence
“accuracy matters”
“consider why it might be different”
how far the answer moved (0 = 365 days, 1 = truth)
0.0
0.2
0.4
0.6
0.8
1.0
no thinking
short
long